

10(a)	(induced) e.m.f. proportional to rate	M1
	of change of (magnetic) flux (linkage)	A1
10(b)	current in primary coil gives rise to <u>magnetic</u> flux	B1
	changing (magnetic) flux in core links with secondary coil	B1
	induced e.m.f. (in secondary coil) causes current in load/resistor	B1
10(c)	correct application of turns ratio: to peak voltage ratio, giving $(V_0 / 220) = (450 / 2700)$ or to r.m.s. voltage ratio, giving $(V_{\text{r.m.s.}} / 156) = (450 / 2700)$	C1
	correct application of $\sqrt{2}$ factor: to peak applied e.m.f., giving $220 / \sqrt{2}$ or to peak output em.f., giving $37 / \sqrt{2}$	C1
	$V_{\text{r.m.s.}} = 26 \text{ V}$	A1